

HDOC Day-8: C Programming Answers

Question 1: Total Surface Area of a Capsule-Shaped Solid

A capsule consists of a cylinder and two hemispheres. The two hemispheres together form a sphere.

Formula: Total Surface Area = $2 \times \pi \times r \times h + 4 \times \pi \times r^2$

```
#include <stdio.h>

int main()
{
    float r, h, area;
    const float PI = 3.14159;

    printf("Enter radius: ");
    scanf("%f", &r);

    printf("Enter cylinder height: ");
    scanf("%f", &h);

    area = 2 * PI * r * h + 4 * PI * r * r;

    printf("Total surface area = %.2f", area);

    return 0;
}
```

Explanation: The curved surface area of the cylinder is $2 \times \pi \times r \times h$, and the surface area of the two hemispheres is equal to the surface area of a sphere, $4 \times \pi \times r^2$. Their sum gives the total surface area.

Question 2: Fourth Angle of a Quadrilateral

The sum of all four interior angles of a quadrilateral is 360 degrees.

Formula: Fourth angle = $360 - (\text{angle1} + \text{angle2} + \text{angle3})$

```
#include <stdio.h>

int main()
{
    float angle1, angle2, angle3, angle4;

    printf("Enter three angles: ");
    scanf("%f %f %f", &angle1, &angle2, &angle3);

    angle4 = 360 - (angle1 + angle2 + angle3);

    printf("Fourth angle = %.2f degrees", angle4);

    return 0;
}
```

Explanation: The program takes three angles as input, adds them, and subtracts their sum from 360 degrees to obtain the fourth angle.